MIL-T-12236A(EL)

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Superseding
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# MILITARY SPECIFICATION
## Test Set, Electrical Power I-199( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers Test Set, Electrical Power, I-199( ). The I-199( ) is used to test dynamotors and hand generators. It includes two metering circuits for measuring input and output voltages, current, and ripple measurements of dynamotors and hand generators. By means of built-in load circuits, the test set can be used to make all tests under various load conditions. The test set includes cord CD-954( ) (dynametor connecting cord), Cord CD-955( ) (dynamotor connecting cord), and cord CD-999( ) (battery connecting cord) (see 6.3).

## 2. APPLICABLE DOCUMENTS

2.1 Documents. The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification

### SPECIFICATIONS

#### FEDERAL

| | |
|---|---|
| T-T-871 | Twine, Cotton, Wrapping |
| NN-P-71 | Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial) |
| QQ-S-781 | Strapping, Steel, Flat and Seals |
| PPP-B-585 | Box, Wood, Wirebound |
| PPP-B-601 | Box, Wood, Cleated-Plywood |
| PPP-B-621 | Box, Wood, Nailed and Lock Corner |

FSC 6625

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| | |
|---|---|
| PPP-B-636 | Box, Shipping, Fiberboard |
| PPP-B-640 | Boxes, Fiberboard, Corrugated, Triple Wall |
| PPP-F-320 | Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes |
| PPP-P-291 | Paperboard, Wrapping, Cushioning |
| PPP-S-760 | Strapping, Nonmetallic, (and Connectors) |
| PPP-T-45 | Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing |
| PPP-T-76 | Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant, (For Carton Sealing) |
| PPP-T-97 | Tape, Pressure-Sensitive Adhesive, Filament Reinforced |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-packaging, Methods of |
| MIL-P-11268 | Parts, Materials, and Processes Used in Electronic Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finish for Ground Signal Equipment |
| MIL-P-19644 | Plastic Molding Material (Polystyrene Foam - Expanded Bead) |
| MIL-P-26514 | Polyurethane Foam, Rigid or Plastic, for Packaging |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-147 | Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-781 | Reliability Tests Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods |



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DRAWINGS

ELECTRONICS COMMAND

| | |
|---|---|
| SC-DL-43593 | Drawing List - Test Set I-199( ) |
| SC-A-46439 | List of Accessories for Package Tester |

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article Samples. When specified in the contract of purchase order, the contractor shall furnish First Article Units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with SC-DL-43593 and the requirements specified herein. Any discrepancies between the drawings and the performance characteristics specified herein, this specification shall have priority.

3.3 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply.

3.4 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be final Film E, semi-gloss light gray enamel, conforming to MIL-F-14072.

3.5 Marking. Marking shall conform to MIL-M-13231.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.5.2 Serial numbers. Serial numbers are required.

3.6 Preconditioning.



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3.6.1 Bounce preconditioning. Each equipment produced and first article shall be subjected to the bounce preconditioning of 4.6.1. After b ounce preconditioning the equipment shall have no loose or broken parts and controls. If a failure occurs, the loose or broken part or control shall be repaired or replaced and the test repeated. Bounce preconditioning shall be performed prior to electrical preconditioning and Group A inspection.

3.6.2 Electrical preconditioning. Each equipment produced and first article shall be be subjected to the electrical preconditioning of 4.6.2. Electrical preconditioning shall be performed after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for requirements of 3.7.3 during period of test.

3.7 Electrical.

3.7.1 Accuracy. The equipment shall meet the accuracy requirements listed below (see 4.7.1).

| MEASUREMENTS | RANGE | ACCURACY |
|---|---|---|
| Resistance | 1000 tc 16250 ohm | ±10% of value |
| DC Current | | |
| Input HIGH | 0-30A and 0-60A | |
| LOW | 0-3A and 0-12A | ±3% of indicated reading |
| Output | 0-120mA and 0-600mA | |
| DC Voltage | | |
| Input | 0-30V ana 0-60V | ±3% of indicated reading |
| Output | 0-300V and 0-1200V | |
| Ripple | | |
| Input | 0-3V and 0-30V | ±5% of indicated reading |
| Output | 0-3V, 0-12V and 0-120V | |

3.7.1.1 Resistance. When tested, the H.V. Load resistance of each test set shall be within the accurancies tested above (see 4.7.1.1).

3.7.1.2 DC Current. When tested, each test set shall measure DC Current to within the accuracies listed above (see 4.7.1.2).



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3.7.1.3 DC Voltage. When tested, each test set shall measure DC Voltage to within the accuracies listed above (see 4.7.1.3).

3.7.1.4 Ripple. When tested, each test set shall measure ripple voltage to within the accuracies stated above (see 4.7.1.4).

3.7.2 Cords. One each of Cord CD-954, Cord CD-955, and Cora CD-999 shall be furnished with each test set. When tested, there shall be no evidence of dielectric breakdown and the insulation resistance shall be greater than 500 megohms (see 4.7.2).

3.7.3 Operational. When specified, the equipment shall be subjected to the following operational requirements.

| FUNCTION | MEASUREMENT | REQ PARA | TEST PARA |
|---|---|---|---|
| RESISTANCE | | 3.7.1.1 | 4.7.1.1 |
| POSITION 1 | INFINITY | | |
| 2 | 14625-17875 ohms | | |
| 3 | 12825-15675 ohms | | |
| 4 | 11025-13475 ohms | | |
| 5 | 9225-11275 ohms | | |
| 6 | 7425-9075 ohms | | |
| 7 | 5625-6875 ohms | | |
| 8 | 3825-4675 ohms | | |
| 9 | 2925-3575 ohms | | |
| RESISTANCE | 900-1000 ohms | | |
| DC CURRENT | | 3.7.1.2 | 4.7.1.2 |
| INPUT | 1.45A-1.55A | | |
| | 2.91A-3.09A | | |
| | 5.82A-6.18A | | |
| | 11.64A-12.36A | | |
| | 14.55A-15.45A | | |
| | 29.10A-30.9A | | |
| | 24.25A-25.75A | | |
| | 48.50A-51.5A | | |
| OUTPUT | 58.2mA-61.8mA | | |
| | 116.4mA-123.6mA | | |
| | 291mA-309mA | | |
| | 582mA-618mA | | |



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| FUNCTION | MEASUREMENT | REQ PARA | TEST PARA |
|---|---|---|---|
| DC VOLTAGE | | 3.7.1.3 | 4.7.1.3 |
| INPUT | 14.55V-15.45V<br>29.1V-30.9V<br>29.1V-30.9V<br>58.2V-61.8V | | |
| OUTPUT | 145.5V-154.5V<br>291V-309V<br>582V-618V<br>1164V-1236V | | |
| RIPPLE VOLTAGE | | 3.7.1.4 | 4.7.1.4 |
| INPUT | 28.5V-31.5V<br>2.85V-3.15V | | |
| OUTPUT | 95V-105V<br>11.4V-12.6V<br>2.85V-3.15V | | |

3.8 Environmental. The Test Set I-199( ) shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following environmental conditions.

3.8.1 Vibration. The test set shall show no evidence of breakage or loosening of parts, and shall meet the electrical requirements of 3.7.3 before and after vibration test, when tested in accordance with 4.8.1.

3.8.2 Bounce, loose cargo. The test set shall show no evidence of physical damage, except minor surface abrasion and shall meet the electrical requirements of 3.7.3 before and after bounce test, when tested in accordance with 4.8.2.

3.8.3 Shock, bench handling. The test set shall show no evidence of physical damage, except minor surface abrasion and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.3.



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3.8.4 Altitude. The test set shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.4.

3.8.5 High temperature. The test set shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.5.

3.8.6 Low temperature. The test shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.6.

3.8.7 Humidity. The test set shall exhibit no physical damage of parts and material, such as corrosion, rust, blistering, swelling, deterioration, etc., or electrical degradation, and shall meet the electrical requirements of 3.7.3 at times specified, when tested in accordance with 4.8.7.

3.8.8 Fungus. The test set shall show no evidence of viable fungus and shall meet the electrical requirements of 3.7.3 before and after fungus test, when tested in accordance with 4.8.8.

3.9 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall meet Requirement 7 of MIL-STD-454. Inspection shall be in accordance with 4.9.

3.10 Workmanship. The test set shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives.
b. Cables, cords, and wires, electrical.
c. Wire (harnessing).
d. Lubricants.
e. Plastic material and parts.
f. Rubber.
g. Tropicalization of material and parts.
h. Wiring and cabling.
i. Connectors (electrical).
j. Screws, other threaded devices, and related parts.
k. Tightening of threaded parts.
l. Riveting.
m. Brazing, soldering, and welding.
n. Welding.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery ) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.11).

4.3 First article. Unless otherwise specified in the contract or purchase order the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish three (3) first article units of the completely assembled Test Set I-199( ).

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, and Group C (see Tables I, II, and III, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units, and designated units for specific subgroups within Group C in accordance with 4.5.3.1. After completion of Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.



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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery, with visual and mechanical inspection in accordance with Table I:

| Item | Where required |
|---|---|
| Parts, materials and processes-general | 3.3 |
| Finish | 3.4 |
| Marking | 3.5 |
| Workmanship | 3.10 |

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4, 4.6, and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government may review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government sampling inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels, and AQL's indicated in Table I. Group A inspection shall be performed in an order satisfactory to the Government.



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Table I - Group A Inspection

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual & mechanical | | | 1 dphu* for the group | 4 dphu for the group |
| Parts, materials and processes-general | 3.3 | 4.10 | | |
| Finish, protective | 3.4 | | | |
| Marking | 3.5 | | | |
| Workmanship | 3.10 | | | |
| Electrical | | | | |
| Resistance | 3.7.1.1 | 4.7.1.1 | 1 dphu for the group | ** |
| DC current | 3.7.1.2 | 4.7.1.2 | | |
| DC voltage | 3.7.1.3 | 4.7.1.3 | | |
| Ripple | 3.7.1.4 | 4.7.1.4 | | |

* Deficiencies per hundred units.
** All electrical deficiencies are major.

4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL's listed in Table II, shall be as follows:

| AQL | Inspection Level |
|---|---|
| 2.0 dphu | S4 |
| 4.0 dphu | S4 |

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.



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Table II - Group B Inspection

| Inspection | Req. Para. | Test Para. | AQL |
|---|---|---|---|
| Interchangeability | 3.9 | 4.9 | 4 dphu* |
| Cords | 3.7.2 | 4.7.2 | 2.0 dphu |

* Defects per hundred units

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met all Group A and B tests. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Three (3) samples of the complete equipment shall be selected at random, one (1) sample for Subgroup 1, one (1) sample for Subgroup 2, and one (1) sample for Subgroup 3, in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirements for first 100 units produced.

4.5.3.1.2 Thereafter, two (2) samples of the complete equipment shall be selected at random, one (1) sample for each Subgroup 1 and 2 in Table III from every 100 units and fraction thereof produced.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order shown in each Subgroup in Table III.



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Table III - Group C Inspection

| Inspection | Req Para. | Test Para. |
|---|---|---|
| Subgroup 1 | | |
| Vibration | 3.8.1 | 4.8.1 |
| Shock, Bench handling | 3.8.3 | 4.8.3 |
| Bounce, loose cargo | 3.8.2 | 4.8.2 |
| Subgroup 2 | | |
| Altitude | 3.8.4 | 4.8.4 |
| High temperature | 3.8.5 | 4.8.5 |
| Low temperature | 3.8.6 | 4.8.6 |
| Humidity | 3.8.7 | 4.8.7 |
| Subgroup 3 | | |
| Fungus | 3.8.8 | 4.8.8 |

4.5.3.3 Evaluation of samples submitted to individual Group C inspection. After samples referenced in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test, unless otherwise specified in the individual tests, the sample shall meet requirements of 3.7.3.

4.5.3.4 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and C sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or purchase order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C samples. Unless otherwise specified, samples which have been subjected to and passed Group C inspection may be delivered on the contract or purchase order provided samples meet requirements of 4.5.4.

4.6 Preconditioning.



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4.6.1 Bounce preconditioning. Each test set produced and first article shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000-SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.6.1.

4.6.2 Electrical preconditioning. Each equipment produced and first article shall be placed in its normal operating position on a test table or in a test rack with room temperature at 77° ±5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period and monitored. During the test, each switch shall be operated, with input voltages changed as requiredfor each range to indicate a meter deflection of 1/2 of the full scale reading. The electrical preconditioning shall be applied uniformly over the 24 hour period for the six (6) functions as given in 3.7.1.2 - 3.7.1.4 to assure preconditioning all controls and parts of the equipment. In case of failure, the equipment shall be repaired, recalibrated, and the electrical preconditioning test repeated for the failed equipment. If required in the contract or purchase order, a failed item analysis report shall be submitted on each failure with contents of report in accordance with 5.10.5 of MIL-STD-781. Each equipment shall meet the requirements of 3.6.2.

4.7 Electrical tests. Unless otherwise specified, test equipment shall be calibrated to an accuracy of at least five times the accuracy of the item under test.

4.7.1 Accuracy test.

4.7.1.1 Resistance.

a. Set OUTPUT switch to RIPPLE, H.V. LOAD FINE ADJ to extreme counterclockwise position. Disconnect any voltage sources from I-199( ).

b. The resistance shall be measured across DYNAMOTOR H.V. terminal for each position of the H.V. LOAD COURSE ADJ. given in 3.7.3.

c. With the H.V. LOAD FINE ADJ control at the extreme clockwise position and the H.V. LOAD COURSE ADJ. in position 9, the resistance shall be between 900 and 1000 ohms.



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d. Inability of the equipment to meet the requirements of 3.7.1.1 shall constitute failure of this test.

4.7.1.2 DC Current.

4.7.1.2.1 DC input current test.

a. Connect a suitable power supply (capable of delivering 3 to 60 amps DC) to the BAT + and DYNAMOTOR LV + terminals of the I-199( ). Set the controls of the I-199( ) as follows:

| CONTROL | SETTING |
|---|---|
| INPUT | AMP |
| INPUT CURRENT | LOW |
| INPUT RHEO | Fully clockwise |

b. Monitor the output of the power supply with a calibrated standard and adjust the power supply until the standard reads 1.5A with the I-199( ) 12 AMPS-3 AMPS switch depressed. If the I-199( ) INPUT meter reads backwards, reverse the connection of the power supply. The INPUT meter of I-199( ) shall read between 1.45A and 1.55A.

c. Adjust the power supply so that the standard reads 3A with the I-199( ) 12 AMPS - 3 AMPS switch depressed. The INPUT meter of I-199( ) shall read between 2.91A and 3.09A.

d. Adjust the power supply so that the standard reads 6A. The I-199( ) 12 AMP - 3 AMP switch is not depressed. The INPUT meter of I-199( ) shall read between 5.82A and 6.18A.

e. Adjust the power supply so that the standard reads 12 A. The I-199( ) 12 AMP - 3 AMP switch is not depressed. The INPUT meter of I-199( ) shall read between 11.64A and 12.36A.

f. Set INPUT CURRENT switch at HIGH. Adjust the power source so that the standard reads 15A with I-199( ) 60 AMP - 30 AMP switch depressed. The INPUT meter of I-199( ) shall read between 14.55A and 15.45A.

g. Adjust the power source so that the standard reads 30 A with the I-199( ) 60 AMP - 30 AMP switch depressed. The INPUT meter of I-199( ) shall read between 29.1A and 30.9A.



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h. With INPUT CURRENT switch on HIGH adjust the power supply so that the standard reads 25 A. Do not depress the I-199( ) 60 AMP - 30 AMP switch. The INPUT meter of I-199( ) shall read between 24.25A and 25.75A.

i. Adjust the power supply so that the standard reads 50A. Do not depress the I-199( ) 60 AMP - 30 AMP switch. The INPUT meter of I-199( ) shall read between 48.5A and 51.5A. CAUTION - Perform this test rapidly since the circuit is fused with a 50A fuse.

4.7.1.2.2 DC output current test.

a. Connect the power supply to the DYNAMOTOR H.V. + and - terminals of I-199( ). Set the OUTPUT switch to MA, the H.V. LOAD COARSE ADJ. to position 9 (extreme clockwise position), and H.V. LOAD FINE ADJ control at the extreme clockwise position. Adjust the power supply controls to deliver 60mA to 600mA.

b. Monitor the output of the power supply with a calibrated standard and adjust the power supply until the standard reads 60mA with the I-199( ) 600mA-120mA switch depressed. The OUTPUT meter of I-199( ) shall read between 58.2mA and 61.8mA.

c. Adjust the power supply so that the standard reads 120mA with the I-199( ) 600MA-120MA switch depressed. The OUTPUT meter of I-199( ) shall read between 116.4mA and 123.6mA.

d. Adjust the power supply so that the stanaard reads 300mA (do not depress the I-199( ) 600MA-120MA switch). The OUTPUT meter of I-199( ) shall read between 291mA and 309mA.

e. Adjust the power supply so that the standard reads 600mA. Do not depress the I-199( ) 600MA-120MA switch. The OUTPUT meter of I-199( ) shall read between 582mA and 618mA.

4.7.1.2.3 Inability of the equipment to meet the requirements of 3.7.1.2 shall constitute failure of this test.

4.7.1.3 DC Voltage.

4.7.1.3.1 DC input voltage test.



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a. Connect the power supply to BAT+ and BAT- of I-199( ). Set power supply controls to deliver 15 VDC to 60 VDC. Set input control switch, SW 1 or S1, to DYN. ON and the INPUT switch to VOLTS. Monitor the output of the power supply with a calibrated standard and adjust the power supply until the standard reads 15 volts with the I-199( ) 60 VOLT - 30 VOLT switch depressed. The INPUT meter of I-199( ) shall read between 14.55V and 15.45V.

b. Adjust the power supply controls so that the standard reads 30V when the I-199( ) 60 VOLT - 30 VOLT switch is depressed The INPUT meter of I-199( ) shall read between 29.1V and 30.9V.

c. Adjust the power supply so that the standard reads 30V. Do not depress the I-199( ) 60 VOLT - 30 VOLT switch. The INPUT meter of I-199( ) shall read between 29.1V and 30.9V.

d. Adjust the power supply so that the standard reads 60V. Do not depress the I-199( ) 60 VOLT - 30 VOLT switch. The INPUT meter of I-199( ) shall read between 58.2V and 61.8V.

4.7.1.3.2 DC Output Voltage Test.

a. Connect power supply to DYNAMOTOR H.V. + and - of I-199( ). Set power supply controls to deliver between 150V and 1200VDC. Set H.V. LOAD COARSE ADJ to OFF and OUTPUT switch to VOLTS. Monitor the output of the power supply with a calibrated standard and adjust the power supply until the standard reads 145V with the I-199( ) 1200 VOLT - 300 VOLT switch depressed. The OUTPUT meter of I-199( ) shall read between 145.5V and 154.5V.

b. Adjust the power supply so that standard reads 300V when the I-199( ) 1200 VOLT - 300 VOLT switch is depressed. The OUTPUT meter of I-199( ) shall read between 291V and 309V.

c. Adjust the power supply so that the standard reads 600V. Do not depress the I-199( ) 1200 VOLT - 300 VOLT switch. The OUTPUT meter of I-199( ) shall read between 582V and 618V.

d. Adjust the power supply so that standard reads 1200V. Do not depress the 1200 VOLT - 300V switch. The OUTPUT meter of I-199( ) shall read between 1164V and 1236V.

4.7.1.3.3 Inability of the equipment to meet the requirements of 3.7.1.3 shall constitute failure of this test.



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4.7.1.4 Ripple.

4.7.1.4.1 Input Ripple Test.

a. Connect a suitable audio oscillator to the HAND GEN L.V. binding posts with the VTVM in parallel.

b. Adjust the test set as follows:

(1) INPUT RHEO and HAN GEN L.V. LOAD fully clockwise.

(2) INPUT CURRENT on LOW.

(3) DYN. ON - OFF - HAND GEN. ON switch on OFF.

(4) INPUT swithc on RIPPLE.

c. Set the audio oscillator to deliver 2000 Hz.

d. Adjust the oscillator output control until the VTVM reads 30V. The I-199( ) INPUT meter shall read between 28.5V and 31.5V.

e. Adjust the oscillator output control until the VTVM reads 3.0V with the I-199( ) RIPPLE 30 VOLT - 3 VOLT switch depressed. The I-199( ) INPUT meter shall read between 2.85V and 3.15V.

4.7.1.4.2 Output ripple voltage test.

a. Connect the audio oscillator to the H.V. DYNAMOTOR binding posts with the VTVM in parallel. (The oscillator output impedance should be between 5000 and 6000 ohms.)

b. Adjust the test set controls as follows.

(1) H.V. LOAD COARSE ADJ and H.V. LOSD FINE ADJ. fully counter-clockwise.

(2) OUTPUT switch on RIPPLE.

c. Set the audio oscillator to deliver 2000 Hz.



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d. Adjust the oscillator output control until the VTVM reads 100V. The I-199( ) OUTPUT meter shall read between 95V and 105V.

e. Adjust the oscillator output control until the VTVM reads 12V with the I-199( ) 12 VOLTS switch depressed. The I-199( ) OUTPUT meter shall read between 11.4V and 12.6V.

f. Adjust the oscillator output control until the VTVM reads 3V with the I-199( ) 3 VOLTS switch depressed. The I-199( ) OUTPUT meter shall read between 2.85V and 3.15V.

4.7.1.4.3 Inability of the equipment to meet the requirements of 3.7.1.4 shall constitute failure of this test.

4.7.2 Cords. All cords shall be inspected and tested for continuity and dielectric strength.

4.7.2.1 Continuity. The cords shall be tested for continuity and correct connections to its terminations, using a low-voltage circuit with a suitable indicating device. Cords indicating open or shorted conditions shall be repaired or replaced, and retested for continuity.

4.7.2.2 Dielectric Strength. A test potential of 2400 volts dc, or peak 60Hz ac shall be applied to each cord for at least 10 seconds. The insulation resistance shall then be measured. Inability of the cord to meet the requirements of 3.7.2 shall constitute failure of this test.

4.8 Environmental. The test set shall be tested in accordance with the following environmental conditions.

4.8.1 Vibration. The test set shall be subjected to the vibration test specified in Part 1, Procedure IX, Method 514 of MIL-STD-810 in the horizontal, parallel to the major horizontal axis of the equipment; horizontal, at right angles to the major horizontal axis of the equipment; horizontal, at right angles to the major horizontal axis of the equipment, and vertically to the major axis of the equipment. Inability of the equipment to meet the requirements of 3.8.1 shall constitute failure of this test.

4.8.2 Bounce, loose cargo. The test set, as prepared for field transportation, shall be subjected to the loose cargo bounce that specified in Part 2, Procedure IX Method 514 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.2 shall constitute failure of this test.



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4.8.3 Shock, bench handling. The test set shall be subjected to the bench handling shock test specified in Procedure V, Method 516 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.3 shall constitute failure of this test.

4.8.4 Altitude. The test set, in its normal operating configuration shall be subjected to the altitude test specified in Procedure 1, Method 500 of MIL-STD-810 with altitude in Step 3 held for 1 hour. Inability of the equipment to meet the requirements of 3.8.4 shall constitute failure of this test.

4.8.5 High temperature. The test set, in its normal operating configuration, shall be subjected to the high temperature test specified in Procedure 1, Method 501 of MIL-STD-810, with temperature in Step 4 held for 12 hours. Inability of the equipment to meet the requirements of 3.8.5 shall constitute failure of this test.

4.8.6 Low temperature. The test set, in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperature in Step 2 held for 48 hours and in Step 4, held for 12 hours. Inability of the equipment to meet the requirements of 3.8.6 shall constitute failure of this test.

4.8.7 Humidity. The test set, in its normal operating configuration, shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810. Inability of the equipment to meet the requirements of 3.8.7 shall constitutes failure of this test.

4.8.8 Fungus. The equipment shall be subjected to the fungus test specified in Method 508, Procedure 1, of MIL-STD-810. There shall be abundant growth colonization (6.6) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.8.8 shall constitute failure of this test.

4.9 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.9. Wher a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.



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d. Location and spacing of test terminals.

4.10 Visual and mechanical. The equipment shall be subjected to a thorough visual and mechanical inspection. The defects shall be classified in accordance with MIL-STD-252, and the documents referenced in requirement paragraphs listed in Table I, Group A inspection.

4.11 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

## 5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Test Set I-199( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Test Set I-199( ) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit package. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Cords, CD-955, CD-999 and CD-954. Each cord shall be packaged method III as follows: Wind each cord into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871.

5.1.1.4.3 Fuses. Fuses for the Test Set I-199( ) shall be packaged method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type I. Secure the wrap with gummed paper tape conforming to PPP-T-45, type III, grade A.

5.1.1.4.4 Consolidated package. Consolidated the items, packaged as specified in 5.1.1.4.2 and 5.1.1.4.3 within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification.



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5.1.1.4.5 Test Set I-199( ). Each test set shall be individually packaged method III as follows: Cushion each set on all surfaces with cells or pads or both conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned item and the consolidated package as specified in 5.1.1.4.4, together within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Fill the void with a dummy fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. Test Set I-199( ) shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents (see 6.2).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of Test Sets I-199( ), packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type, PPP-B-621, style 4, class 2, or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.



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5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Test Sets I-199( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Test Set I-199( ), packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively, selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Test Set I-199( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Test Set I-199( ), packaged as specified in 5.1, shall be packed as specified in 5.2.2.2, except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Test Set I-199( ) is intended for use at operator/crew level.

6.2 Ordering data. Procurement documents should specify the following

a. Title, number, and date of this specification and any amendment thereto.



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b. Level of preservation, packaging, packing and marking (see Section 5).

c. If rough handling and functional tests are required.

d. Quantity of first articles required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example. SG-12( )/U to SG-12F/U.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.4).

6.5 Sampling inspection. Sampling by the Government will be limited to the amount deemed necessary to determine compliance with the contract or purchase order and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of sampling inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Definitions.

6.6.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.6.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together, or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.



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6.6.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.6.4 Sporulation. Sproulation is the formation of minute unicellular reproductive or dormont bodies, called spores.

6.6.5 Tubular germination. Tuburiai germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-12236A(EL) TEST SET, ELECTRICAL POWER I-199( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

ESC-FM 1068-68

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